



Aror University of Art, Architecture, Design & Heritage Sukkur

Faculty of Emerging Sciences & Technologies (FEST) BS (Artificial Intelligence) CS-207 Database Systems

Course Title: Database Systems
Course Code: CS-207
Credit Hours: 4 (3+1)
Course Instructor: Engr. Muhammad Younis
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Course Introduction:

The course aims to introduce basic database concepts, different data models, data storage and retrieval techniques and database design techniques. The course primarily focuses on relational data model and DBMS concepts.

Assessment:

S. #	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes & Assignments	30%	5
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs) :

CLO #	Course Learning Outcomes	Bloom Taxonomy	Mapped (GAs)
CLO-1	Explain fundamental database concepts.	C2 (Explain)	GA1
CLO-2	Design conceptual, logical, and physical database schemas using different data models.	C5 (Design)	GA3
CLO-3	Identify functional dependencies and resolve database anomalies by normalizing database tables.	C2 (Identify)	GA2
CLO-4	Use Structured Query Language (SQL) for database definition and manipulation in any DBMS.	C4 (Use)	GA5

Reference Books

1. Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition by Thomas Connolly and Carolyn Begg
2. Database Systems: The Complete Book, 2nd Edition by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom

3. Database System Concepts, 6th Edition by Avi Silberschatz, Henry F. Korth and S. Sudarshan.
4. Database Management Systems, 3rd Edition by Raghu Ramakrishnan, Johannes Gehrke

Course Outlines:

Week #	Lec: No.	Topic	Reference	% of Covered Course
1	1	Basic Database Concepts	Notes / Slides	2.38
	2	Database Approach vs File-Based System	Notes / Slides	4.76
	3	Advantages of DBMS	Notes / Slides	7.14
2	4	Database Architecture	Notes / Slides	9.52
	5	Three-Level Schema Architecture	Notes / Slides	11.90
	6	Data Independence	Notes / Slides	14.29
3	7	Introduction to SQL	Notes / Slides	16.67
	8	SQL Data Definition Language (DDL)	Notes / Slides	19.05
	9	SQL Data Manipulation Language (DML)	Notes / Slides	21.43
4	10	SQL Joins	Notes / Slides	23.81
	11	SQL Subqueries & Nested Queries	Notes / Slides	26.19
	12	SQL Constraints	Notes / Slides	28.57
5	13	Grouping in SQL	Notes / Slides	30.95
	14	Aggregation Functions in SQL	Notes / Slides	33.33
	15	HAVING Clause	Notes / Slides	35.71
6	16	Relational Data Model	Notes / Slides	38.10
	17	Attributes, Schemas, Tuples, Domains	Notes / Slides	40.48
	18	Relation Instances	Notes / Slides	42.86
7	19	Keys of Relations	Notes / Slides	45.24
	20	Integrity Constraints	Notes / Slides	47.62
	21	Introduction to Relational Algebra	Notes / Slides	50.00
8	–	Mid Term Examination	–	–
9	22	Relational Algebra: Selection	Notes / Slides	52.38
	23	Relational Algebra: Projection	Notes / Slides	54.76
	24	Cartesian Product & Types of Joins	Notes / Slides	57.14
10	25	Functional Dependencies	Notes / Slides	59.52
	26	Normalization (1NF, 2NF)	Notes / Slides	61.90
	27	Normalization (3NF)	Notes / Slides	64.29
11	28	Boyce-Codd Normal Form (BCNF)	Notes / Slides	66.67
	29	Higher Normal Forms	Notes / Slides	69.05
	30	Normalization Problem Solving	Notes / Slides	71.43
12	31	Entity Relationship Model	Notes / Slides	73.81
	32	Entity Sets, Attributes, Relationships	Notes / Slides	76.19

Week #	Lec: No.	Topic	Reference	% of Covered Course
	33	Entity Relationship Diagrams (ERD)	Notes / Slides	78.57
13	34	Concurrency Control	Notes / Slides	80.95
	35	Transactions & ACID Properties	Notes / Slides	83.33
	36	Deadlocks & Serializability	Notes / Slides	85.71
14	37	Database Backup	Notes / Slides	88.10
	38	Database Recovery Techniques	Notes / Slides	90.48
	39	Indexes	Notes / Slides	92.86
15	40	Introduction to NoSQL Systems	Notes / Slides	95.24
	41	Types of NoSQL Databases	Notes / Slides	97.62
	42	SQL vs NoSQL	Notes / Slides	100.00
16	–	Final Examination	–	–

GA: Graduate Attributes

GA1 Computing Knowledge: An ability to apply knowledge of mathematics, science, computing fundamentals and computing specialization to the solution of complex computing problems.

GA2 Problem Analysis: An ability to identify, formulate, research literature, and analyze complex computing problems reaching substantiated conclusions using first principles of mathematics, natural sciences and computing sciences.

GA3 Design/Development of Solutions: An ability to design solutions for complex computing problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

GA4 Investigation: An ability to investigate complex computing problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

GA5 Modern Tool Usage: An ability to create, select and apply appropriate techniques, resources, and modern IT tools, including prediction and modeling, to complex computing activities, with an understanding of the limitations.

GA6 The Computer Scientist and Society: An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional computing practice and solution to complex computing problems.

GA7 Environment and Sustainability: An ability to understand the impact of professional computing solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

GA8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.

GA9 Individual and Team Work: An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

GA10 Communication: An ability to communicate effectively, orally as well as in writing, on complex computing activities with the computing community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

GA11 Project Management: An ability to demonstrate management skills and apply computing principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

GA12 Lifelong Learning: An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments